

Solution: $A \oplus B = (A \cup B) - (A \cap B)$ (Prob 41)

Let, the i th bit string of A and B be $(0,0)$. $\therefore$ i th bit of $A \cup B = 0$ and $A \cap B = 0$. By Prob 61, i th bit of $A \oplus B$ is 0 .

Let, the i th bit string of A and B be $(0,1)$. $\therefore$ i th bit of $A \cap B = 0$ and $A \cup B = 1$. By Prob 61, i th bit of $A \oplus B$ is 1 .

Let, the i th bit string of A and B be $(1,0)$. $\therefore$ i th bit of $A \cup B = 1$ and $A \cap B = 0$. By Prob 61, i th bit of $A \oplus B$ is 1 .

Let, the i th bit string of A and B be $(1,1)$. $\therefore$ i th bit of $A \cup B = 1$ and $A \cap B = 1$. By Prob 61, i th bit of $A \oplus B$ is 0 . $\square$

The successor of the set A is the set $A \cup \{A\}$.

65) Find the successors of the following sets

(a) $\{1,2,3\}$: $\{1,2,3\} \cup \{\{1,2,3\}\} = \{1,2,3, \{1,2,3\}\}$

(b) $\emptyset$: $\emptyset \cup \{\emptyset\} = \{\emptyset\}$ (Commutative Law 1)

(c) $\{\emptyset\}$: $\{\emptyset\} \cup \{\{\emptyset\}\} = \{\emptyset, \{\emptyset\}\}$ (Identity Law 2)

(d) $\{\emptyset, \{\emptyset\}\}$: $\{\emptyset, \{\emptyset\}\} \cup \{\{\emptyset, \{\emptyset\}\}\} = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$

66) How many elements does the successor of a set with n elements have?

Solution: WLOG, $A = \{a_1, a_2, \dots, a_n\}$. Successor of $A = \{a_1, a_2, \dots, a_n\} \cup \{A\} = \{a_1, a_2, \dots, a_n, \{a_1, a_2, \dots, a_n\}\}$

$\therefore$ |Successor(A)| = $n+1$. $\square$

The Jaccard similarity $J(A,B)$ of the finite sets ~~A and B~~ A and B is $J(A,B) = \frac{|A \cap B|}{|A \cup B|}$, with $J(\emptyset, \emptyset) = 1$. The Jaccard distance $d_J(A,B)$ between A and B equals $d_J(A,B) = 1 - J(A,B)$.

71) Find $J(A,B)$ and $d_J(A,B)$ for these pairs of sets.

(a) $A = \{1,3,5\}$, $B = \{2,4,6\}$ $A \cup B = \{1,2,3,4,5,6\}$ $|A \cup B| = 6$ $A \cap B = \emptyset$
 $|A \cap B| = 0$ $J(A,B) = \frac{0}{6} = 0$ $d_J(A,B) = 1 - 0 = 1$